

The background of the slide features a complex network diagram. It consists of numerous nodes of varying sizes and colors (dark blue, light blue, and grey) connected by thin, light grey lines. Some nodes are highlighted with larger, concentric circles. The overall aesthetic is modern and technological.

K-NEAREST NEIGHBOR ALGORITHM

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AGENDA

Introduction

KKN Algorithm

Example

Advantages

Disadvantages

INTRODUCTION

What is KNN?

K-Nearest Neighbors (KNN) is one of the simplest machine learning algorithms. It's used in **supervised learning**, which means it needs **labeled data** to learn.

KNN can be used for:

- **Classification** – predicting categories (e.g., spam or not spam)
- **Regression** – predicting continuous values (e.g., house prices)

□ Simple Idea:

When we want to predict something new, we just **look at the K most similar examples** from the past and **make a decision based on them**.

For example:

If you want to know the fruit you're holding, and 3 people around you say it's an "apple", you'd likely call it an apple too.

KKN ALGORITHM

K-nearest neighbors (KNN) algorithm uses 'feature similarity' to predict the values of new data points which further means that new data point will be assigned a value on how closely it matches the points in the training set. We can understand its working with the help of following steps

Step 1- For implementing any algorithm, we need dataset. So during the first step of KNN, we must load the training as well as test data.

Step 2- Next, we need to choose the value of K i.e. the nearest data points. K can be any integer.

Step 3- For each point in the test data do the following

3.1 Calculate the distance between SlidePlayer 4/9 test data and each row of training data with the help of any of the method namely: the Euclidean, Manhattan or Hamming distance. based The most commonly used method to calculate distance is Euclidean.

3.2 Now, based on the distance value, sort them in ascending order.

3.3 Next, it will choose the top K rows from the sorted array.

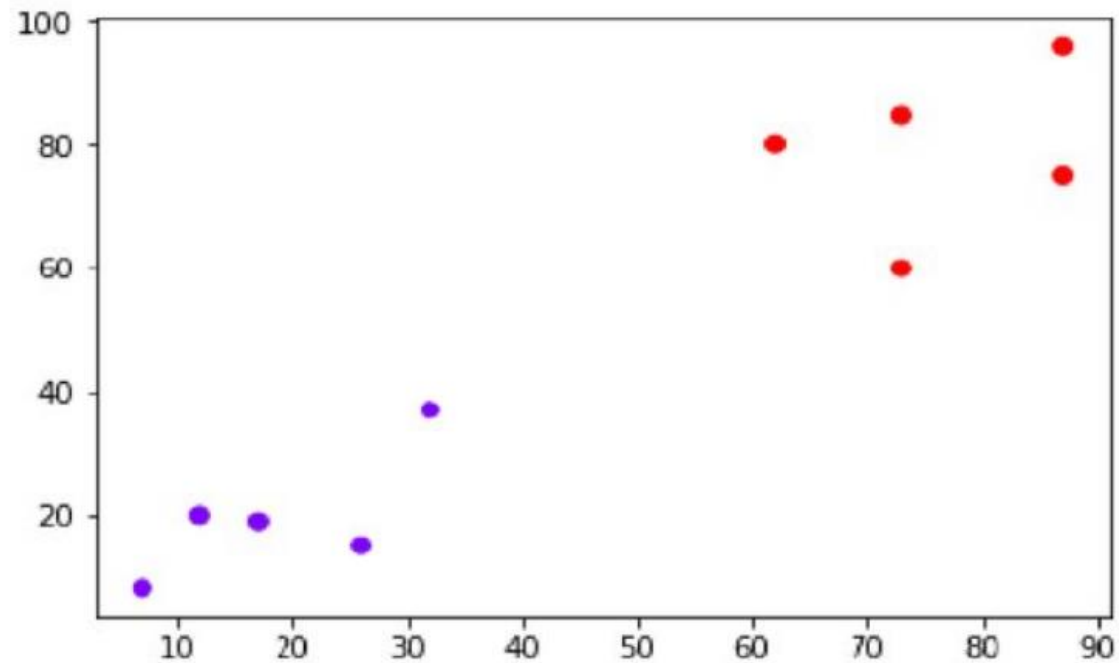
3.4 Now, it will assign a class to the test point based on most frequent class of these row

Step 4 - End

EXAMPLE

The following is an example to understand the concept of K and working of KNN algorithm

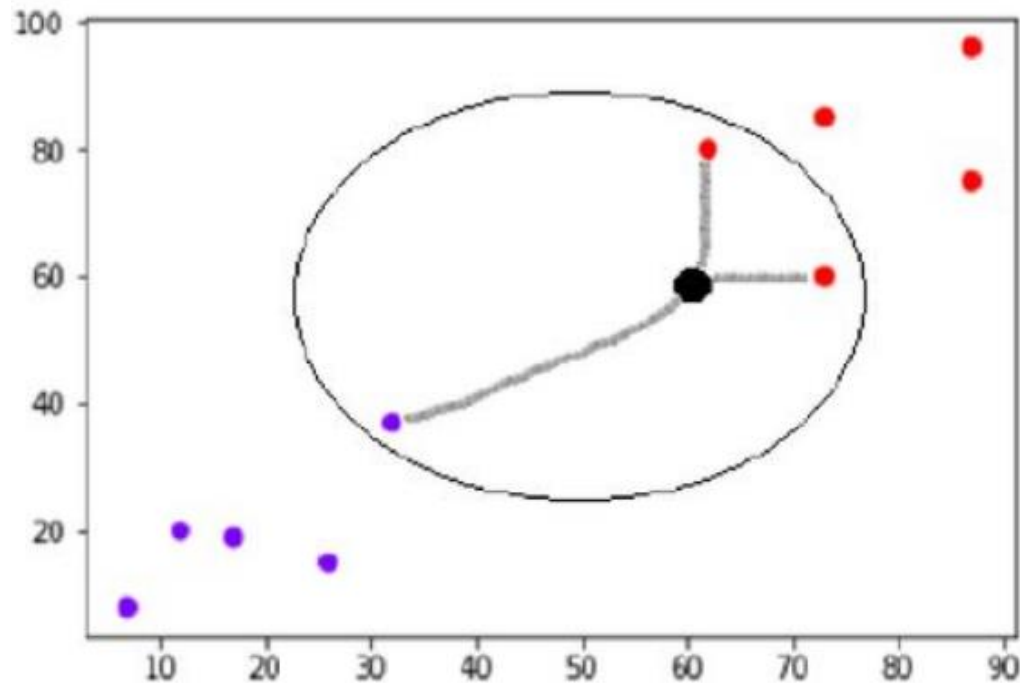
Suppose we have a dataset which can be plotted as follows:



Example (Conti..)

Now, we need to classify new data point with black dot (at point 60,60) into **blue** or **red** class. We are assuming $K = 3$ i.e. it would find three nearest data points.

It is shown in the following diagram:



We can see in the beside diagram the three nearest neighbors of the data point with **black** dot. Among those three, two of them lies in **Red** class hence the black dot will also be assigned in red class.

ADVANTAGES (PROS)

- ✓ **Simple and Easy to Understand**

KNN doesn't involve complex calculations or training phases.

- ✓ **No Training Time**

It stores the data and waits for a query – which makes it a **lazy learner**.

- ✓ **Handles Multi-Class Classification Well**

You can classify into more than 2 categories easily.

- ✓ **Adaptable to Non-linear Data**

Can work even if your data isn't linearly separable.

(like **Logistic Regression** or **Linear SVM**)

DISADVANTAGES (CONS)

✗ Slow Prediction Time

Because KNN compares with every data point, it becomes **slow with large datasets**.

✗ Sensitive to Irrelevant Features

It treats all features equally. If some are not useful, it still considers them.

(KNN gives **equal importance to all features**, whether they are helpful or not.)

✗ Needs Feature Scaling

If features are on different scales, results will be misleading.

(Without scaling, KNN might think that two people are very different just because one earns more — even if they're the same age and category.)

✗ Struggles with High Dimensions

When you have lots of features (high-dimensional data), distances become less meaningful — known as the **curse of dimensionality**.

K-NN ALGORITHM MATHEMATICALLY

K-Nearest Neighbors Algorithm

BMI	Age	Sugar
33.6	50	1
26.6	30	0
23.4	40	0
43.1	67	0
35.3	23	1
35.9	67	1
36.7	45	1
25.7	46	0
23.3	29	0
31	56	1

- Apply K nearest neighbor classifier to predict the diabetic patient with the given features BMI, Age. If the training examples are,
- Assume $K=3$,
- Test Example BMI=43.6, Age=40, Sugar=?

BMI	Age	Sugar	Distance	
33.6	50	1	$\sqrt{(43.6 - 33.6)^2 + (40 - 50)^2}$	14.14
26.6	30	0	$\sqrt{(43.6 - 26.6)^2 + (40 - 30)^2}$	19.72
23.4	40	0	$\sqrt{(43.6 - 23.4)^2 + (40 - 40)^2}$	20.20
43.1	67	0	$\sqrt{(43.6 - 43.1)^2 + (40 - 67)^2}$	27.00
35.3	23	1	$\sqrt{(43.6 - 35.3)^2 + (40 - 23)^2}$	18.92
35.9	67	1	$\sqrt{(43.6 - 35.9)^2 + (40 - 67)^2}$	28.08
36.7	45	1	$\sqrt{(43.6 - 36.7)^2 + (40 - 45)^2}$	8.52
25.7	46	0	$\sqrt{(43.6 - 25.7)^2 + (40 - 46)^2}$	18.88
23.3	29	0	$\sqrt{(43.6 - 23.3)^2 + (40 - 29)^2}$	23.09
31	56	1	$\sqrt{(43.6 - 31)^2 + (40 - 56)^2}$	20.37

